

HR Employee Attrition Prediction

Introduction

Employee attrition is one of the biggest challenges faced by organizations. Attrition means employees leaving the company due to resignation, retirement, dissatisfaction, or better opportunities. High attrition can increase recruitment costs, training expenses, and reduce overall productivity.

This project focuses on analyzing employee data and predicting whether an employee is likely to leave the organization. By using data analytics and machine learning techniques, companies can identify the main reasons for attrition and take preventive actions to improve employee retention.

Objectives

The objectives of this project are:

- To analyze employee data and identify attrition trends
 - To understand factors affecting employee turnover
 - To predict whether an employee may leave the company
 - To help organizations improve employee retention strategies
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Dataset Description

The dataset used for this project is the **IBM HR Analytics Employee Attrition Dataset** obtained from Kaggle.

Dataset Features:

- Total Records: 1470 Employees
- Total Columns: 35 Features

Important Columns:

- Age
- Department
- Job Role
- Monthly Income
- Years At Company
- Job Satisfaction
- Work Life Balance
- Overtime

- Attrition (Yes / No)
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Tools & Technologies Used

The following tools were used in this project:

- **Google Colab** – Code execution platform
 - **Python** – Programming language
 - **Pandas** – Data analysis
 - **NumPy** – Numerical operations
 - **Matplotlib / Seaborn** – Data visualization
 - **Scikit-learn** – Machine learning model building
 - **GitHub** – Project hosting and version control
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Methodology

The project was completed in the following stages:

Step 1: Data Collection

The dataset was downloaded from Kaggle and uploaded to Google Colab.

Step 2: Data Cleaning

- Checked missing values
- Removed duplicate entries
- Converted categorical values into numerical form

Step 3: Exploratory Data Analysis

Visualizations were created to understand:

- Attrition count
- Department-wise attrition
- Salary vs attrition
- Age distribution
- Overtime impact

Step 4: Feature Selection

Important columns selected:

- Age
- Monthly Income

- Years At Company
- Overtime

Step 5: Model Building

Logistic Regression model was used to predict attrition.

Step 6: Model Evaluation

Accuracy score and confusion matrix were used to evaluate performance.

6 Data Visualization & Analysis

📌 Attrition Count

The number of employees who left the company was lower than employees who stayed.

📌 Department-wise Attrition

Some departments showed higher attrition rates compared to others.

📌 Income vs Attrition

Employees with lower monthly income were more likely to leave.

📌 Overtime Impact

Employees working overtime had a higher chance of attrition.

7 Machine Learning Model

Model Used:

- Logistic Regression

Why Logistic Regression?

It is simple, efficient, and suitable for binary classification problems like:

- Leave = Yes
 - Leave = No
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8 Results

The machine learning model successfully predicted employee attrition with good accuracy.

Key Findings:

- Low salary increases attrition risk
- Overtime employees are more likely to leave

- Employees with fewer years in company leave more
 - Certain departments have higher attrition
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9 Business Recommendations

Based on the analysis, organizations should:

- Improve employee salary packages
 - Reduce overtime workload
 - Increase employee engagement
 - Provide career growth opportunities
 - Improve work-life balance
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10 Advantages of This Project

- Helps HR team take early action
 - Reduces hiring and training costs
 - Improves employee satisfaction
 - Supports data-driven decision making
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1 1 Conclusion

This project successfully analyzed employee attrition using data analytics and machine learning techniques. The model identified key factors influencing attrition and predicted which employees are likely to leave.

The project demonstrates how companies can use employee data to improve retention strategies and build a more stable workforce.

1 2 Future Scope

- Use advanced models like Random Forest and XGBoost
 - Build real-time HR dashboard
 - Add employee feedback sentiment analysis
 - Deploy as web application
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1 3 References

- IBM HR Analytics Dataset – Kaggle

- [Python Documentation](#)
- [Scikit-learn Documentation](#)
- [Google Colab](#)